Cross-supplier LFP state-of-health forecasting using a PyBaMM-trained neural operator

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The reuse of retired lithium iron phosphate (LFP) cells is limited by the time and cost required to determine their future degradation behaviour. Conventional lifetime assessment requires extended cycling, while purely empirical models often transfer poorly between cells with different designs and manufacturing histories. We present a physics-conditioned neural operator that forecasts the state-of-health (SoH) trajectory of second-life LFP cells from a compact diagnostic characterisation.

The workflow first estimates cell-specific electrochemical parameters from short rest-voltage, pulse-response and partial capacity measurements (approximate open-circuit voltage inversion, pulse-derived series resistance, and an initial-capacity anchor). These parameters condition a Neural ODE [2] that predicts SoH as a function of cycle number. The operator is trained using 490 degradation trajectories generated in PyBaMM [1] by perturbing five parameters governing solid-electrolyte-interphase (SEI) growth (rate, molar volume, solvent diffusivity), lithium plating, and negative-electrode loss of active material, around seven experimentally characterised anchor cells drawn from three commercial LFP suppliers (CALB, EVE, REPT). A censored-likelihood objective allows trajectories that do not reach the selected end-of-life threshold within the simulation horizon to contribute to training.

Evaluation was performed on three held-out experimental cells, one per supplier, using only the nearest-supplier-anchor parameter vector as a prior (no per-cell degradation-curve fit). The model achieved in-window SoH root-mean-square errors of 0.53, 1.10 and 0.97 percentage points on the actual (nameplate-normalised) SoH scale for the CALB, EVE and REPT cells respectively (Figure 1). The standard deviation of the cell-level errors was 0.24 percentage points, providing preliminary evidence of consistent performance across the three evaluated commercial LFP cell families. We hypothesise that conditioning on physically interpretable parameters, rather than raw characterisation traces, contributes to the observed transferability.

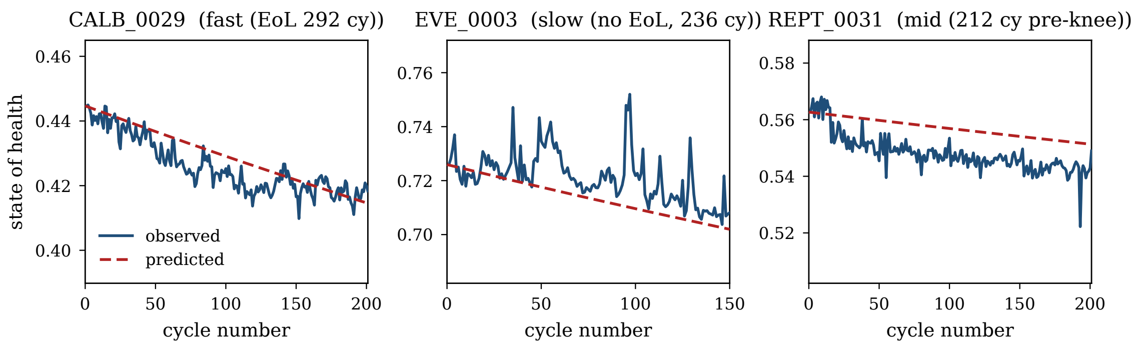

Figure 1: Observed vs forecast SoH on three held-out cells (one per supplier). SoH is expressed as discharge capacity divided by nameplate capacity, so second-life cells enter the plot below 1.0. The operator forecast (red dashed) is anchored at the observed first-cycle SoH; no cell-specific degradation fit is used. In-window SoH RMSE: 0.53, 1.10, 0.97 percentage points; cell-level std = 0.24 pp.

The proposed framework connects short experimental diagnostics, PyBaMM-generated lifetime simulations and computationally efficient trajectory forecasting. Ongoing work will expand the number of experimental anchor and validation cells, quantify forecast uncertainty, and evaluate end-of-life-cycle prediction under application-specific second-life thresholds.

References:

1. V. Sulzer, S. G. Marquis, R. Timms, M. Robinson, S. J. Chapman, “Python Battery Mathematical Modelling (PyBaMM)”, *Journal of Open Research Software* **9** (2021), 14.